Supplementary Material

A Bayesian Joint Longitudinal–Interval and Multi-State Threshold-Crossing Framework for Irregular Molecular Monitoring in Chronic Myeloid Leukemia

This supplement supports the main paper with: (i) the closed-form running extrema and the optional relapse frailty of the multi-state threshold-crossing model (Section 3.3 of the main paper); (ii) an exploratory informative visit-process model; and (iii) two secondary simulation scenarios (endpoint choice and informative monitoring). The multi-state model, its simulation recovery, and its application to the cohort are reported in the main paper.

S1 Closed-form running extrema and relapse frailty

Because the latent trajectory $m_i(\ell)$ is quadratic in $\ell = \log(1+t)$ (piecewise-quadratic once a spline slope change is added), the running extrema that define the multi-state transitions are available in closed form. The running minimum equals the value at the vertex $\ell^* = -(\beta_1 + b_{1i})/(2\beta_2)$ clamped to the observed range $[0, \ell(t)]$; past the vertex m_i is monotone increasing, so the upward (relapse) crossing and the durability maximum are point evaluations. This removes the soft-minimum grid approximation entirely and yields a smooth log density, which in practice eliminates the divergent transitions the grid induces.

Between-patient heterogeneity in relapse propensity can be captured by a Gaussian frailty $\rho_i \sim N(0, \sigma_\rho^2)$ on the relapse threshold. Because

$$\int \Phi\left\{\frac{x+\rho}{\sigma_{\text{thr}}}\right\} \mathrm{d}N(\rho; 0, \sigma_\rho^2) = \Phi\left\{\frac{x}{\sqrt{\sigma_{\text{thr}}^2 + \sigma_\rho^2}}\right\},$$

the frailty is equivalent to a wider relapse-specific width $\sigma_{\text{rel}} = \sqrt{\sigma_{\text{thr}}^2 + \sigma_\rho^2}$ and preserves the closed form. In the cohort the frailty modestly improved relapse calibration but was only weakly identified (9 confirmed relapses) and added little over the spline model in simulations where trajectory curvature already explains relapse; the base spline model is therefore preferred.

S2 Exploratory informative visit-process model (not fitted)

Monitoring intensity may itself depend on molecular response: a patient doing well may be seen less often, and more frequent testing yields more opportunities to document DMR. This dependence is written into the governing guideline, which raises molecular-monitoring frequency when the BCR-ABL transcript rises [1]. To assess sensitivity to informative visiting, one may model the visit intensity jointly with the trajectory,

$$\lambda_i^V(t) = \exp\{\delta_0 + \delta_1[m_i(t) + 4.5] + \delta_2 \log(1+t) + u_i\}, \qquad u_i \sim N(0, \sigma_u^2),$$

so that visit intensity depends on the latent MRD, time, and a patient-level monitoring random effect u_i. For visit times $v_{i1}, \ldots, v_{iM_i}$ in $[E_i, C_i]$ the visit-process likelihood is

$$L_i^V = \left\{ \prod_{\ell=1}^{M_i} \lambda_i^V(v_{i\ell}) \right\} \exp\left\{ -\int_{E_i}^{C_i} \lambda_i^V(s)\, ds \right\},$$

with the cumulative intensity evaluated by quadrature. Because this extension adds weakly identified parameters in a cohort of 87 patients, it is presented as a sensitivity model to be evaluated primarily by simulation and in larger cohorts.

S3 Simulation: endpoint choice and informative monitoring

The main-text simulation focuses on the primary joint model and the multi-state recovery study. Here we report the two secondary scenarios summarised in the patient-level calibration table (Table 12 of the main paper).

Endpoint choice. A latent threshold-crossing estimator, which targets biological onset rather than the visit-documented endpoint, under-predicted documented DMR more strongly than the floor model (mean predicted ≈ 0.44 versus observed ≈ 0.85 at $n = 150$; difference ≈ -0.41, versus ≈ -0.26 for the floor model). This quantifies the distinction between the latent-crossing and documented-DMR estimands: because documentation occurs only at visits, an estimator that targets biological onset necessarily predicts fewer *documented* events under interval ascertainment.

Informative monitoring. Under a response-adaptive visit process the observed documented-DMR rate fell (to ≈ 0.59) and the floor model's calibration slope rose above one (1.16), showing that the monitoring process materially affects calibration. This motivates the joint visit-process extension outlined as future work in the main paper.

S4 Simulation-based calibration

Figure S1 shows the posterior rank histograms from the simulation-based calibration of Section 6.2 of the main paper (150 prior-predictive draws at $n = 87$; ranks of the true value among 100 thinned posterior draws). Under a correctly calibrated posterior the ranks are Uniform; bars within the grey 95% band are consistent with uniformity. Ten of the eleven parameters are consistent with uniformity; only the intercept β_0 shows a mild departure, within multiple-testing expectation across eleven parameters.

References

[1] Chinese Society of Hematology, Chinese Medical Association. Chinese guidelines for the diagnosis and treatment of chronic myeloid leukemia (2011 edition). *Chinese Journal of Hematology (Zhonghua Xue Ye Xue Za Zhi)*, 32(6):426–432, 2011.

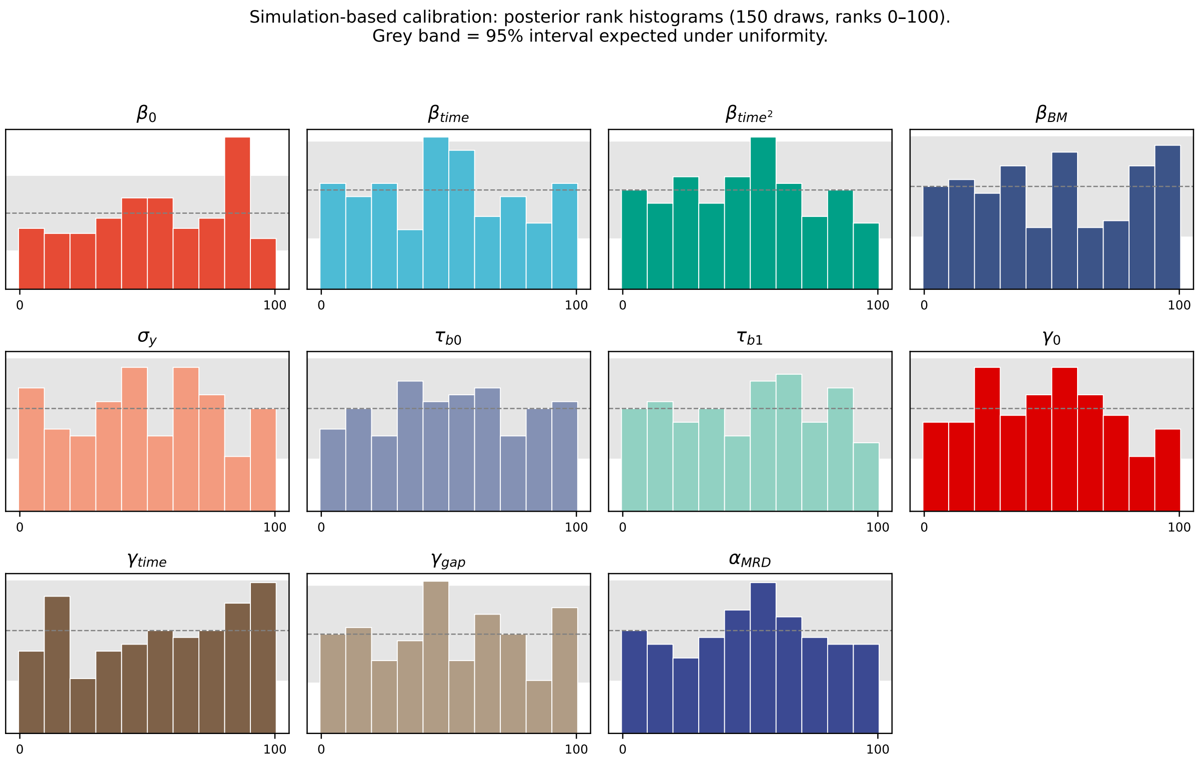

Figure S1: Simulation-based calibration: posterior rank histograms for each parameter (150 draws). Grey band = 95% interval expected under uniformity; dashed line = expected count.